Theory of Value

An Axiomatic Analysis of Economic Equilibrium
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Chapter 1: Mathematics

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Functions

Let S and T be two sets; if with each element $x \in S$ is associated one and only one element $y \in T$, a function f from S to T is defined. f is also called a transformation of S into T. x is the variable, y is the image of x by f, or the transform of x by f, or the value of f at x, and one writes $y = f(x)$ (read y equals f of x), or $x \to f(x)$.

Example: Let A be the set of actions available to a certain agent. If the choice of a in A determines the amount of money (a positive integral number of cents) this agent receives, a function from A to $\mathbb{N}$ is defined.

Functions

Consider in $S \times T$ the set of elements (x, y) for which $y = f(x)$. This subset of $S \times T$ is called the graph of the function f.

$$G(f) = \{(x, y) \in S \times T \mid y = f(x)\}.$$

Example: The graph of a function f from S to T is visualized as a set of points of the rectangle such that the vertical through an arbitrary point x of S intersects it at exactly one point.

Functions

Let X be a subset of S; take the image $y = f(x)$ of each $x \in X$. The set of the images so obtained is a subset of T called the image of X and denoted $f(X)$.

$$f(X) = \{f(x) \in T \mid x \in X\}.$$

If $f(S)$ consists of one element of T, in other words, if all the elements of S have the same image in T, the function f is said to be constant. If $f(S) = T$, in other words, if each element of T is the image of some element of S, f is said to be a function from S onto T, i.e., function $f : S \to T$ is a surjective (onto) mapping.

For any $y \in T$, there exists $x \in S$ such that $f(x) = y$.

Functions

Theorem 1. If f is a function from a set S to a set T, and if $\mathscr{X}$ is a collection of subsets of S, then

(1) $f\left(\bigcup_{X\in\mathscr{X}} X\right) = \bigcup_{X\in\mathscr{X}} f(X)$.

(2) $f\left(\bigcap_{X\in\mathscr{X}} X\right) \subseteq \bigcap_{X\in\mathscr{X}} f(X)$.

Remark: There exist a function f from a set S to a set T, and $\mathscr{X}$ is a collection of subsets of S, such that then

$f\left(\bigcap_{X\in\mathscr{X}} X\right) \neq \bigcap_{X\in\mathscr{X}} f(X)$.

Example: Let $A = \{1, 2\}$, $f(1) = f(2) = 0$, then $f(\{1\}) \cap f(\{2\}) = \{0\}$ while $\{1\} \cap \{2\} = \varnothing$.

Functions

Theorem 1. If f is a function from a set S to a set T, and if $\{X_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of S, then

(1) $f\left(\bigcup_{\lambda\in\Lambda} X_\lambda\right) = \bigcup_{\lambda\in\Lambda} f(X_\lambda)$.

(2) $f\left(\bigcap_{\lambda\in\Lambda} X_\lambda\right) \subseteq \bigcap_{\lambda\in\Lambda} f(X_\lambda)$.

Remark: There exist a function f from a set S to a set T, and $\{X_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of S, such that then $f\left(\bigcap_{\lambda\in\Lambda} X_\lambda\right) \neq \bigcap_{\lambda\in\Lambda} f(X_\lambda)$.

Example: Let $A = \{1, 2\}$, $f(1) = f(2) = 0$, then $f(\{1\}) \cap f(\{2\}) = \{0\}$ while $\{1\} \cap \{2\} = \varnothing$.

Functions

Conversely, let Y be a subset of T; the set of $x \in S$ which have their images in Y is a subset of S called the inverse image of Y and denoted $f^{-1}(Y)$.

$$f^{-1}(Y) = \{x \in S \mid f(x) \in Y\}.$$

Functions

Theorem 2. If f is a function from a set S to a set T, and if $\mathscr{Y}$ is a collection of subsets of T, then It is easy to prove that:

(1) $f^{-1}\left(\bigcup_{Y\in\mathscr{Y}} Y\right) = \bigcup_{Y\in\mathscr{Y}} f^{-1}(Y)$.

(2) $f^{-1}\left(\bigcap_{Y\in\mathscr{Y}} Y\right) = \bigcap_{Y\in\mathscr{Y}} f^{-1}(Y)$.

(3) $f^{-1}(Y \setminus Y') = f^{-1}(Y) \setminus f^{-1}(Y')$ for $Y' \subseteq Y \subseteq T$.

Functions

Theorem 2. If f is a function from a set S to a set T, and if $\{Y_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of T, then It is easy to prove that:

(1) $f^{-1}\left(\bigcup_{\lambda\in\Lambda} Y_\lambda\right) = \bigcup_{\lambda\in\Lambda} f^{-1}(Y_\lambda)$.

(2) $f^{-1}\left(\bigcap_{\lambda\in\Lambda} Y_\lambda\right) = \bigcap_{\lambda\in\Lambda} f^{-1}(Y_\lambda)$.

(3) $f^{-1}(Y \setminus Y') = f^{-1}(Y) \setminus f^{-1}(Y')$ for $Y' \subseteq Y \subseteq T$.

Functions

A $y \in T$ may be the image of several, or of one, or of no element of S. When for each $y \in T$ the set $f^{-1}(y)$ consists of exactly one element, in other words, when each $y \in T$ is associated with one and only one element of S, f is said to establish a one-to-one correspondence between S and T, i.e., function $f : S \to T$ is an injective (one-to-one) mapping.

For any $x_1 \in S$ and $x_2 \in S$, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

For any $x_1 \in S$ and $x_2 \in S$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

Functions

Let f be a function from a set F to a set T, and let G be a set containing F. A function g from G to T is said to be an extension of f to G if one has $f(x) = g(x)$, for every x in F.

Let f be a function from a set F to a set T, and let G be a set contained in F. A function g from G to T is said to be an restriction of f to G if one has $f(x) = g(x)$, for every x in G.

Functions

Consider N sets $S_1, \cdots, S_n, \cdots, S_N$ and their product $\prod_{n=1}^{N} S_n$. The function which associates with the generic element (x_n) of $\prod_{n=1}^{N} S_n$, its nth coordinate x_n in S_n is called the nth projection (or the projection on S_n). The image of an element (resp. of a set) by a projection function is called the projection of that element (resp. of that set).

Correspondences

If with each element x in a given set S is associated a non-empty subset Y of a given set T, a function φ, from S to the set of subsets of T is defined. It is sometimes preferable to consider φ as a correspondence from S to T. One writes $Y = \varphi(x)$.

Example: (1) Let f be a function from a set S onto a set T; for every $y \in T$, $f^{-1}(y)$ is a non-empty subset of S, hence f^{-1} is a correspondence from T to S. (2) Let A be the set of a priori available actions of a social agent. Suppose that his environment is completely specified by an element e of a set E. His environment restricts his freedom of action, i.e., the element e determines the subset of A to which his choice is actually restricted. A correspondence from E to A is thus introduced.

Correspondences

Let S and T be two sets. If with each element x of S we associate a subset $\varphi(x)$ of T, we say that the correspondence $x \longrightarrow \varphi(x)$ is a mapping of S into T; the set $\varphi(x)$ is called the image of x under the mapping φ. The set $S^* = \{x \in S \mid \varphi(x) \neq \varnothing\}$ is called the domain (or set of definition) of φ and $T^* = \bigcup_{x \in S} \varphi(x)$ is called the range (or set of values) of φ; we also say that φ is defined on S^* and that it is a mapping of S^* onto T^*, i.e., correspondence $\varphi : S^* \to T^*$ is a surjective (onto) mapping.

If the mapping φ of S into T is such that the set $\varphi(x)$ always consists of a single element, we say that φ is a single-valued function or a single-valued mapping of S into T.

Correspondences

A mapping is called semi-single-valued if

$$\varphi(x) \cap \varphi\left(x'\right) \neq \varnothing \Longrightarrow \varphi(x) = \varphi\left(x'\right).$$

Clearly a single-valued mapping is also semi-single-valued. A mapping is called injective (one-to-one) if

$$x \neq x' \Longrightarrow \varphi(x) \cap \varphi\left(x'\right) = \varnothing.$$

An injective mapping is evidently semi-single-valued.

If φ is a mapping of S into T and A is a non-empty subset of S, we write

$$\varphi(A) = \bigcup_{x \in A} \varphi(x).$$

If $A = \varnothing$, we write $\varphi(\varnothing) = \varnothing$. The set $\varphi(A)$ is called the image of A under the mapping φ; if we compare x to a source of light, making a shadow $\varphi(x)$ on a screen, then $\varphi(A)$ is the shadow produced by a set A of sources of light.

If $\mathscr{A} = \{A_\lambda \mid \lambda \in \Lambda\}$ is a family of sets, we write

$$\varphi(\mathscr{A}) = \{\varphi\left(A_\lambda\right) \mid \lambda \in \Lambda\}$$

and call $\varphi(\mathscr{A})$ the image of $\mathscr{A}$ under the mapping φ.

Correspondences

Theorem 3. $A \subseteq A'$ implies that $\varphi(A) \subseteq \varphi(A')$.

Theorem 4. $\varphi\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) = \bigcup_{\lambda\in\Lambda} \varphi\left(A_\lambda\right)$.

Theorem 5. $\varphi\left(\bigcap_{\lambda\in\Lambda} A_\lambda\right) \subseteq \bigcap_{\lambda\in\Lambda} \varphi\left(A_\lambda\right)$.

Theorem 6. If $A \subseteq S$ and if φ is a mapping of S with range T, then

$$\mathbf{C}\left(\varphi(A)\right) \subseteq \varphi(\mathbf{C}A).$$

If, further, φ is injective, then

$$\mathbf{C}\left(\varphi(A)\right) = \varphi(\mathbf{C}A).$$

Correspondences

Proof of Theorem 6. $y \in \mathbf{C}(\varphi(A))$, $y \notin \varphi(A) = \bigcup_{x \in A} \varphi(x)$, there isn't $x \in A$ such that $y \in \varphi(x)$, then $y \in \varphi(x)$ for some $x \in \mathbf{C}A$, and hence $y \in \varphi(\mathbf{C}A)$.

Conversely, $y \in \varphi(\mathbf{C}A) = \bigcup_{x \in \mathbf{C}A} \varphi(x)$, then $y \in \varphi(x)$ for some $x \in \mathbf{C}A$. If φ is injective, then $y \notin \varphi(x)$ for $x \in A$, and $y \in \varphi(x)$ for $x \notin A$, thus there isn't $x \in A$ such that $y \in \varphi(x)$, $y \notin \bigcup_{x \in A} \varphi(x) = \varphi(A)$, $y \in \mathbf{C}(\varphi(A))$.

Correspondences

The graph of the correspondence φ is a subset of $S \times T$, namely

$$G(\varphi) = \{(x, y) \in S \times T \mid y \in \varphi(x)\}.$$

Example: Consider the case of the rectangle already used to illustrate the graph of a mapping: the graph of correspondence from S to T is visualized as a set of points of the rectangle: the intersection of this set with a vertical through an arbitrary point x of S is a non-empty set (the projection of which on T is $\varphi(x)$).

Correspondences

A binary operation $\top$ on a set S associates with each pair (x, y) of elements of S (the order of which is essential) a unique element z of S. One writes $z = x \top y$. Thus a binary operation on S is nothing else than a function from $S \times S$ to S.

Example: (1) The addition of two positive integers is a binary operation $+$ on $\mathbb{N}$. (2) So is the multiplication of two positive integers. (3) Given two positive integers x, y, the expression x^y note the product of y positive integers equal to x; thus a binary operation $\wedge$ on $\mathbb{N}$ can be defined by $x \wedge y = x^y$.

Correspondences

A binary operation $\top$ on S is said to be associative if for all x, y, z in S one has $(x \top y) \top z = x \top (y \top z)$. It is said to be commutative if for all x, y in S one has $x \top y = y \top x$.

Example: (1) Both $+$ and $\cdot$ on $\mathbb{N}$ are associative and commutative. (2) $\wedge$ on $\mathbb{N}$ is neither associative nor commutative.

Correspondences

Consider two binary operations $\perp$ and $\top$ defined on S. The first is said to be distributive with respect to the second if for all x, y, z in S one has $x \perp (y \top z) = (x \perp y) \top (x \perp z)$.

Example: (1) On $\mathbb{N}$, $\cdot$ is distributive with respect to $+$. (2) But $+$ is not distributive with respect to $\cdot$.

Operations on mappings

If φ_1 and φ_2 are two mappings of S into T, their union is the mapping $(\varphi_1 \cup \varphi_2)$ of S into T defined by

$$(\varphi_1 \cup \varphi_2)(x) = \varphi_1(x) \cup \varphi_2(x).$$

The intersection of φ_1 and φ_2 is the mapping $(\varphi_1 \cap \varphi_2)$ of S into T defined by

$$(\varphi_1 \cap \varphi_2)(x) = \varphi_1(x) \cap \varphi_2(x).$$

The Cartesian product of φ_1 and φ_2 is the mapping $(\varphi_1 \times \varphi_2)$ of S into $T \times T$ defined by

$$(\varphi_1 \times \varphi_2)(x) = \varphi_1(x) \times \varphi_2(x).$$

More generally, given any operations $\vee$ and $\wedge$ on sets, we write

$$(\varphi_1 \vee \varphi_2)(x) = \varphi_1(x) \vee \varphi_2(x);$$
$$(\varphi_1 \wedge \varphi_2)(x) = \varphi_1(x) \wedge \varphi_2(x).$$

Operations on mappings

If φ_1 is a mapping of S into T and φ_2 is a mapping of T into U, the composition product of φ_2 by φ_1 is the mapping $(\varphi_2 \cdot \varphi_1)$ of S into U defined by

$$(\varphi_2 \cdot \varphi_1)(x) = \varphi_2(\varphi_1(x)).$$

We have

(1) $(\varphi_1 \cup \varphi_2)(A) = \varphi_1(A) \cup \varphi_2(A)$;

(2) $(\varphi_1 \cap \varphi_2)(A) \subseteq \varphi_1(A) \cap \varphi_2(A)$;

(3) $(\varphi_1 \times \varphi_2)(A) \subseteq \varphi_1(A) \times \varphi_2(A)$;

(4) $(\varphi_2 \cdot \varphi_1)(A) = \varphi_2(\varphi_1(A))$.

Operations on mappings

A mapping Δ is said to be constant if there exists a subset C of T such that $\Delta(x) = C$ for all S. A constant mapping satisfies the following property:

$$(\Delta \cap \varphi)(A) = \Delta(A) \cap \varphi(A).$$

Another important mapping is the identity mapping, which is the single-valued mapping I of S onto S defined by

$$I(x) = \{x\}.$$

$$\begin{aligned}
(\Delta \cap \varphi)(A) &= \bigcup_{x \in A} (\Delta \cap \varphi)(x) = \bigcup_{x \in A} [\Delta(x) \cap \varphi(x)] \\
&= \bigcup_{x \in A} [C \cap \varphi(x)] = C \cap \left[\bigcup_{x \in A} \varphi(x) \right] \\
&= C \cap \varphi(A) = \Delta(A) \cap \varphi(A).
\end{aligned}$$

Operations on mappings

If $\varphi_1, \varphi_2, \varphi_3$ are mappings of S into S, we have associativity of composition product

$$\varphi_3 \cdot (\varphi_2 \cdot \varphi_1)(x) = (\varphi_3 \cdot \varphi_2) \cdot \varphi_1(x).$$

Thus, if φ is a mapping of S into S, we can write

$$\varphi^2(x) = \varphi(\varphi(x))$$
$$\varphi^3(x) = \varphi\left(\varphi^2(x)\right) = \varphi^2(\varphi(x)), \text{ etc.}$$

The transitive closure of φ is a mapping $\hat{\varphi}$ of S into S defined by

$$\hat{\varphi}(x) = \{x\} \cup \varphi(x) \cup \varphi^2(x) \cup \varphi^3(x) \cup \cdots$$

The correspondence $A \longrightarrow \hat{\varphi}(A)$ is a closure operation, for

(1) $\hat{\varphi}(A) \supseteq A$,

(2) $A \supseteq A' \Longrightarrow \hat{\varphi}(A) \supseteq \hat{\varphi}(A')$,

(3) $\hat{\varphi}(\hat{\varphi}(A)) = \hat{\varphi}(A)$.

Operations on mappings

Theorem 7. If Γ_1 and Γ_2 are two semi-single-valued mappings, $\Gamma_1 \cap \Gamma_2$ and $\Gamma_1 \times \Gamma_2$ are semi-single-valued mappings.

Theorem 8. If one of the mappings Γ_1 and Γ_2 is injective, the mappings $\Gamma_1 \cap \Gamma_2$ and $\Gamma_1 \times \Gamma_2$ are injective.

Upper and Lower Inverses

If φ is a mapping of S into T, its lower inverse is the mapping φ^- of T into S defined by

$$\varphi^-(y) = \{x \in S \mid y \in \varphi(x)\}.$$

This is a mapping whose domain is T^* and whose range is S^*.

Theorem 9. If φ is single-valued, φ^- is injective; if φ is injective, φ^- is single-valued; if φ is semi-single-valued, φ^- is semi-single-valued.

Proof. For $y \neq y'$ we want to prove $\varphi^-(y) \cap \varphi^-(y') = \varnothing$. Otherwise, there exists $x \in \varphi^-(y) \cap \varphi^-(y')$, then $y \in \varphi(x)$ and $y' \in \varphi(x)$. If φ is single-valued, then $y = \varphi(x) = y'$, which is a contradiction.

If $\varphi^-(y) \cap \varphi^-(y') \neq \varnothing$, then there exists $x \in \varphi^-(y) \cap \varphi^-(y')$, then $y \in \varphi(x)$ and $y' \in \varphi(x)$. If φ is injective, then $y = y'$ and hence $\varphi^-(y) = \varphi^-(y')$.

Upper and Lower Inverses

For any non-empty subset B of T we have

$$\varphi^{-}(B) = \{x \in S \mid \varphi(x) \cap B \neq \varnothing\}$$

and we also write $\varphi^{-}(\varnothing) = \varnothing$. It is clear that the inverse of φ^{-} is $(\varphi^{-})^{-} = \varphi$ and that $y \in \varphi(x)$ is equivalent to $x \in \varphi^{-}(y)$.

$$\begin{aligned}
(\varphi^{-})^{-}(A) &= \{y \in T \mid \varphi^{-}(y) \cap A \neq \varnothing\} \\
&= \{y \in T \mid \{x \in S \mid y \in \varphi(x)\} \cap A \neq \varnothing\} \\
&= \{y \in T \mid \{x \in A \mid y \in \varphi(x)\} \neq \varnothing\} \\
&= \{y \in T \mid y \in \varphi(A)\} = \varphi(A)
\end{aligned}$$

$$(\varphi^{-})^{-}(x) = \{y \in T \mid x \in \varphi^{-}(y)\} = \{y \in T \mid y \in \varphi(x)\} = \varphi(x)$$

Upper and Lower Inverses

Apart from the mapping φ^-, it is sometimes useful to consider another correspondence of a similar kind, which we call the upper inverse of φ; in this mapping to each subset B of T there corresponds the set

$$\varphi^+(B) = \{x \in S^* \mid \varphi(x) \subseteq B\}.$$

In particular, if $B = \varnothing$, we have $\varphi^+(\varnothing) = \varnothing$. We observe that for all subsets B we have $\varphi^+(B) \subseteq \varphi^-(B)$.

Upper and Lower Inverses

Theorem 10. If B_1 and B_2 are subsets of T, then we have

(1) $\mathbf{C}\varphi^+(B) = \varphi^-(\mathbf{C}B)$ and $\mathbf{C}\varphi^-(B) = \varphi^+(\mathbf{C}B)$.

(2) $\varphi^+(B_1) \cup \varphi^+(B_2) \subseteq \varphi^+(B_1 \cup B_2)$.

(3) $\varphi^-(B_1) \cup \varphi^-(B_2) = \varphi^-(B_1 \cup B_2)$.

Proof.

$$\begin{aligned} \mathbf{C}\varphi^+(B) &= \mathbf{C}\{x \in S \mid \varphi(x) \subseteq B\} \\ &= \mathbf{C}\{x \in S \mid \varphi(x) \cap \mathbf{C}B = \varnothing\} \\ &= \{x \in S \mid \varphi(x) \cap \mathbf{C}B \neq \varnothing\} = \varphi^-(\mathbf{C}B) \end{aligned}$$

$$\begin{aligned} \mathbf{C}\varphi^-(B) &= \mathbf{C}\{x \in S \mid \varphi(x) \cap B \neq \varnothing\} \\ &= \{x \in S \mid \varphi(x) \cap B = \varnothing\} \\ &= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}B\} = \varphi^+(\mathbf{C}B) \end{aligned}$$

Upper and Lower Inverses

$$\begin{aligned}
& \varphi^+(B_1) \cup \varphi^+(B_2) \\
= \; & \{x \in S \mid \varphi(x) \subseteq B_1\} \cup \{x \in S \mid \varphi(x) \subseteq B_2\} \\
\subseteq \; & \{x \in S \mid \varphi(x) \subseteq B_1 \cup B_2\} \\
= \; & \varphi^+(B_1 \cup B_2)
\end{aligned}$$

$$\begin{aligned}
& \varphi^-(B_1) \cup \varphi^-(B_2) \\
= \; & \{x \in S \mid \varphi(x) \cap B_1 \neq \varnothing\} \cup \{x \in S \mid \varphi(x) \cap B_2 \neq \varnothing\} \\
= \; & \{x \in S \mid (\varphi(x) \cap B_1) \cup (\varphi(x) \cap B_2) \neq \varnothing\} \\
= \; & \{x \in S \mid \varphi(x) \cap (B_1 \cup B_2) \neq \varnothing\} \\
= \; & \varphi^-(B_1 \cup B_2)
\end{aligned}$$

Countable Sets

A sequence $\left(x^1, x^2, \cdots, x^q, \cdots\right)$ of elements of a set S is an intuitive concept. Precisely, it is defined as a function from $\mathbb{N}$ to S. It will he denoted by (x^q).

Countable Sets

Intuitively, a set S is countable if it has at most as many elements as $\mathbb{N}$. In a precise fashion, a set S is defined as countable if it can be put in one-to-one correspondence with a subset of $\mathbb{N}$. When the countable set S is not empty one can always choose the corresponding subset of $\mathbb{N}$ so that it owns 1, and so that, whenever it owns two positive integers, it owns every positive integer between them. The image of $x \in S$ in the correspondence is then called the rank of x. Thus a set is countable if and only if all its elements can be ranked, no two different elements having the same rank. One can prove that

Theorem 11. The product of finite countable sets is countable.

Example: (1) $\mathbb{N} \times \mathbb{N}$ is countable. (2) However, one can prove that the set of subsets $\mathbb{N}$ of is not countable.

Countable Sets

Theorem 12. The set of subsets $\mathbb{N}$ of is not countable.

Proof: We show that there is no surjection from a set A to its power set 2^A. Consider any function $f : A \to 2^A$ and let

$$B = \{a \in A \mid a \notin f(a)\}$$

We claim that there is no $b \in A$ such that $f(b) = B$. Indeed, assume $f(b) = B$ for some $b \in A$. Then either $b \in B$ hence $b \notin f(b)$ which is a contradiction, or $b \notin B$ implying that $b \in B$ which is again a contradiction. Hence the map f is not surjective as claimed. $\square$